

Where and Why is My Bot Failing? A Visual Analytics Approach for Investigating Failures in Chatbot Conversation Flows

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ABSTRACT

The ongoing coronavirus pandemic has accelerated the adoption of AI-powered task-oriented chatbots by businesses and healthcare organizations. Despite advancements in chatbot platforms, implementing a successful and effective bot is still challenging and requires a lot of manual work. There is a strong need for tools to help conversation analysts quickly identify problem areas and, consequently, introduce changes to chatbot design. We present a visual analytics approach and tool for conversation analysts to identify and assess common patterns of failure in conversation flows. We focus on two key capabilities: path flow analysis and root cause analysis. Interim evaluation results from applying our tool in real-world customer production projects are presented.

Index Terms: visual analytics; conversation analytics; visualization systems and tools; chatbots; dialogue systems; root cause analysis

1 INTRODUCTION

During the global coronavirus pandemic, enterprise call centers were under huge pressure to respond to a growing demand of customer care service requests. The travel and workplace restrictions, as well as workforce costs, placed limits on the feasibility of scaling human-staffed call centers to meet demand. In response to these pressures, enterprises accelerated their investments and adoption of AI-based conversational agents (aka chatbots) to scale the support for a wide range of tasks, such as responding to COVID-related questions in healthcare or fulfilling customer service requests in retail.

Unlike traditional software development, the task of developing a task-oriented AI-based chatbot is more complex and includes several additional challenges. Grudin and Jaques [3] describe several of these challenges, including the complexity of the conversation, the generation and understanding of natural language, and the need to constantly adapt to new or unanticipated user input. They conclude that the most important lesson could be that the release of a chatbot is the beginning and not the end of development, and that teams need to plan for continuous improvement. Best practices in the area of continuous improvement, such as those published by Watson Assistant [5], describe key activities along a Measure-Assess-Improve-Test-Deploy lifecycle. However, some of these activities, such as the assessment of failures, still require a lot of manual work.

The conversation analyst is a relatively new role that has emerged in AI-based chatbot development projects. This paper focuses on the analyst's role and presents two complementary visual analytics capabilities that help the analyst discover and assess groups of conversations with similar failure patterns such as leading to abandonment or escalation. These capabilities are designed within a visual analytics paradigm that puts the human at the center of information discovery and analytical processing.

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The main contributions of this paper are: (1) description of the conversation analyst user task and information needs for identifying and assessing groups of problematic conversations; (2) presentation of novel conversation root cause analysis and path flow analysis that we developed and deployed to address these needs; and (3) demonstration of insights and interim evaluation from applying these capabilities in large-scale, real-world production chatbot systems. The rest of the paper is organized as follows: **Section 2** presents the related prior art. **Section 3** describes the role of the conversation analyst and her relevant assessment tasks. **Section 4** presents the root cause analysis and path flow analysis capabilities. **Section 5** describes a couple of projects and datasets in which we deployed our tool. **Section 6** demonstrates initial results from one of these projects, and **Section 7** discusses results and next steps.

2 RELATED ART

Although research exists on detection and exploration of failed chatbot conversations, we are unaware of comprehensive studies that cover all or most essential aspects of the problem. Sandbank et al. [16] detect egregious or extraordinarily bad conversations using an SVM prediction model, based on different features such as user rephrasing, chatbot repetitions, and sentiment analysis. Meena et al. [12] search for misunderstood user utterances that implied wrong dialog system response. The authors do not directly address the success and failure of conversations. Instead, they explore the understanding of single turns. Kvale et al. [11] manually analyze several hundred chatbot conversations and discuss characteristics and root causes of issues in the successful and failed ones.

Academic research on visualization approaches to the exploration of failed conversation outcomes seems to be scarce. Kim et al. [10] provide a review of visualization and summarization methods for human-to-human online conversations. However, human-to-chatbot interactions are significantly different. Rozga [15] provides a survey of two conversational analytics platforms, Chatbase and Dashbot. Chatbase [2] session flow diagram has similarities with ours, but does not provide rich interactive filtering, analysis, and drill-downs.

Both web analytics [18] and business process mining [1] study multi-step processes with potential positive and negative outcomes that are often represented by graphical flow models. In spite of the differences between the domains, exploring common topics can be insightful. Qafari and van der Aalst [14] and Suriadi et al. [17] present different approaches to root cause analysis in process mining.

3 METHODOLOGY AND USER TASK

Commercial AI-based conversational bot platforms [2], [8], [13] include a natural language understanding (NLU) component for detecting user intents and entities¹ from utterances, and a dialog management component to control the dialog logic and the bot responses. For example, the dialog skill in Watson Assistant [7] combines training of the NLU by providing examples, and visual programming of the dialog behavior.

The conversation analyst (hereafter referred to as the analyst) is a multidisciplinary specialist role that combines good understanding

¹Intents represents user questions, and entities represent parameters

of the domain area of the bot, knowledge in conversation design principles, and know-how concerning technical implementations. The analyst is typically involved in all development phases of the chatbot, and often leads the continuous improvement activities due to her intimate familiarity with the expected behavior and responses of the bot. A common practice for analysts is to select a sample of failed conversations and manually assess every conversation failure, labeling incorrect responses, identifying the root cause, and documenting the required fix. However, this is labor intensive and may not necessarily find the biggest issues to solve. Thus, there is a need for more effective tools to identify the biggest issues and facilitate faster assessment of root causes and potential fixes.

The approach and tool described in this paper focus on improving the effectiveness of the analyst in identifying and assessing large groups of conversations that potentially fail for a common reason. Figure 1 depicts the main assessment activities that are supported by our tool and their relationship to the Measure and Improve lifecycle phases. The assessment is typically driven by some improvement goal, e.g., improving the successful conversation rate in a given topic area. The assessment activity includes the following main steps: **Step 1 - Filter and focus** includes interactive UI controls that allow the analyst to search for patterns in specific topic areas, time ranges, or problem areas. **Step 2 - Discover failing patterns** includes interactive analysis and visualization capabilities for identifying groups of similar failing conversations and their corresponding volume and potential improvement impact. **Step 3 - Assess issues in a selected group of conversations** includes additional descriptive analytics and interactive drill-downs that allow analysts to assess the issues in the detected group in more detail and determine the nature of the issue, as well as the corrective action. These steps are not always performed in sequential order. Sometimes the analyst will gain knowledge when observing a detected pattern in Step 2, or during detailed assessment in Step 3, and will go back to Step 1 to refine the filters and focus on a modified scope. From this perspective, our tool follows the visual analytics sense-making loop for extracting knowledge [9].

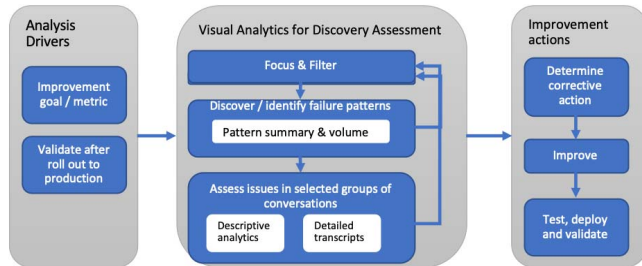


Figure 1: The visual analytics tasks in the assessment phase and their relationship to the Measure and Improve phases.

The tool described in this paper includes two main capabilities that follow this assessment activity from two different angles: (1) conversation flow analysis for understanding conversation flow paths, and for identifying sequence patterns in conversation steps leading to failure; and (2) conversation root cause analysis for identifying significant data features correlated with failure.

4 TOOL DESIGN AND IMPLEMENTATION

The assessment capabilities described in this paper have been developed and deployed as part of a larger conversation analytics system developed in IBM. The overall system includes layers for ingestion of chatbot production logs, a metric computation framework, and a monitoring dashboard for tracking key business and performance indicators. The system also includes service layers for both online synchronous queries and analytics tasks, as well as an offline asynchronous layer for long-running analytics tasks. The root cause analysis and flow analysis capabilities were implemented using the

online synchronous layer to facilitate the few-seconds response time for interactive analysis and visualization needs.

4.1 Root Cause Analysis

The root cause analysis is an interactive visual analytics capability for identifying significant data features and values in conversation logs that are statistically correlated with failure. The goal of this capability is to help the analyst identify where and why conversations are failing. The analyst typically observes in the monitoring dashboard that some failure is happening more than expected, and the root cause analysis is then used to understand the source and reason of these failures in more detail.

Figure 2 presents a screenshot of the root cause analysis user interface (UI). The left pane includes UI controls to allow the analyst to filter and focus on specific date ranges, topic areas, failure types, and features. The right pane includes a visual representation of the most significant data features and values that were detected to be associated with the selected type of failure, and their relative importance and impact on the number of failures. The UI navigation flow starts with updating the filter and focus controls in the left pane, clicking the submit button, and after a few seconds observing the discovered features and values in the right pane. At that point, the analyst can focus on a particular feature by clicking on the relevant row and navigate to another page to assess the conversations associated with that feature values. The analyst can also update the parameters in the left pane and resubmit the analysis.

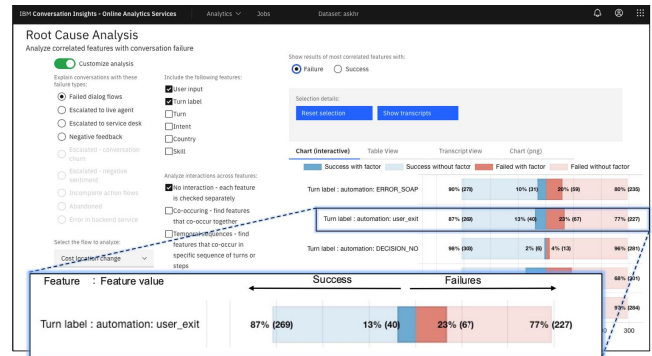


Figure 2: Root cause analysis UI. The visualization of significant feature values shows the magnitude of failures and successes

The root cause analysis algorithm requires two main types of inputs: a group of failed conversations and a group of successful ones. The analyst can choose the failure group by selecting the type of failure. The algorithm then automatically selects the corresponding successful group to be used for analysis. For example, when exploring negative user feedback, conversations with positive user feedback are naturally defined as the successful ones. If conversations with an uncompleted tasks are explored, conversations with the same completed task are used as the comparison group. Additional inputs for the algorithm include selection of the conversation steps or features to take into account for the analysis. For example, the analyst may selectively choose that only the first user utterance or last user utterance will be taken into consideration. Additional options determine which features will be taken into consideration for the analysis, such as *Country*, *User input*, and *Turn label*. These types of selections can improve the quality and accuracy of the results for a specific use case or hypothesis exploration.

Internally, the algorithm transforms the root cause analysis problem into a binary classification (prediction) problem and uses a machine learning model to learn the importance of different features of the problem. Currently, Random Forest is used as a prediction model. A significant advantage of our technique is the combined use of textual features (user inputs, chatbot responses), user-related context variables (e.g., country), and dialog-related variables (e.g.,

intent, dialog node). The algorithm includes several feature engineering capabilities to support these different features and data types. For example, when the analyst selects the *User input* feature, then unigrams and bigrams from the user input will be used as feature values in the model and will be presented in the results. In the case of *Country*, the name of countries will be used as feature values.

The output of the algorithm is a sorted list of the most significant feature values and interactions² that are strongly correlated with failure (or success) and, therefore, distinguish between the two groups of conversations. The visual representation we chose is a centrally aligned stacked horizontal bar chart. The (vertical) X-axis contains the prioritized list of feature values. Four data points are presented for each bar: (1) total failed with this value; (2) total failed in dataset; (3) total success with this value; and (4) total success in the dataset. Red and blue divergent colors are used to represent failures and successes, respectively. This choice of chart type and colors were made to allow the analyst to judge both the absolute and relative magnitudes of the failures. The success bars are mainly designed to enhance explainability and increase trust in the analysis.

4.2 Path Flow Analysis

The conversation path flow analysis (hereafter referred to as flow analysis) is an interactive analysis and visualization tool for understanding user conversation flows within the chatbot dialog system. One of the main use cases is to help the analyst find high volumes of problematic conversation patterns that lead to failures such as user abandonment. Another use case is to observe and validate expected conversation behavior changes after rolling out a new version into production.

The dialog skill in Watson Assistant provides a visual environment for programming how the chatbot should respond to different user inputs and dialog steps to complete the conversation or handle errors. The programming model consists of nodes that get triggered by evaluating condition expressions over user inputs such as intents, entities, or other state variables. Upon triggering, a node can produce a response, wait for user input, or pass the control to another node. A key goal of the flow analysis is to unlock the execution behavior of the chatbot across dialog nodes and help the analyst find sequential patterns along the execution of failed conversations.

To better understand what constitutes a problematic execution pattern of many conversations, we need to explain the characteristics and data of a single conversation. The production logs from Watson Assistant include metadata that details which dialog nodes were actively used (fired) in generating the chatbot response to the user utterance. The data is a list of dialog nodes in every turn. By collecting all turn-level logs of a single conversation, we can reconstruct the full conversation flow. Additional metadata in the logs can help to determine conversation outcomes such as escalation, completion, or abandonment. Therefore, problematic execution patterns are execution flows that are common to many conversations and end up with a negative outcome.

Based on discussions we held with analysts, we developed a visual metaphor for conversation flows that puts extra emphasis on the sequential steps of the conversation. The benefit of sequential aggregation is the ease of finding sequential conversation patterns. The downside of this approach is that aggregating by sequence is sensitive to the sequence variance. The analyst can choose the level of granularity to be used to represent conversation steps in the visual aggregate: (1) **node-level** - each node triggered will be represented in the aggregated flow graph; (2) **turn-level** - the most meaningful node in each conversation turn will be used; or (3) **milestone-level** - only important nodes that are labeled as milestones will be used.

Figure 3 presents a screenshot of the flow analysis UI. Similar to the aforementioned root cause analysis, the UI is organized with

a left pane used for filtering and configuring the algorithm options, and a right pane for the interactive visualization of conversation flows. The visual metaphor is of a tree that represents the possible sequences discovered from the data. Every node in the tree represents a step in a specific conversation path. Every edge represents the sequence between two steps. The sequence is ordered from left to right. Nodes and edges carry additional statistical information and trends, such as the volume of conversation visits in a node, the distribution of the next step in a sequence via the edges, and the ratio of abandonment of every sequence pattern (represented as a red sink). The analyst will typically interact with this chart by navigating the flows with the highest volume and biggest abandonment or other negative outcome. Clicking on a node expands the next level of children nodes and selects the node for additional detail and drill-down purposes, such as exploring other properties and transcripts of conversations that were abandoned at the selected node.

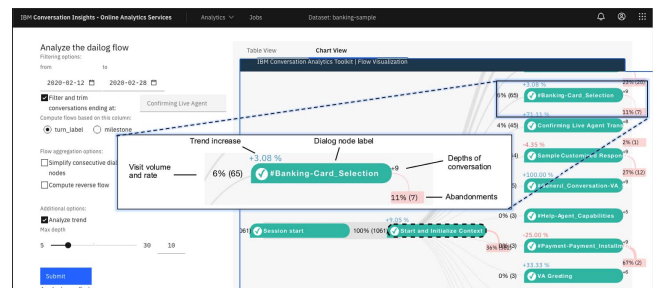


Figure 3: Flow analysis UI. Statistical information such as visits, abandonment and trend over time is presented for each dialog node

In addition, the following features are supported by flow analysis: (1) **Filtering** - users can select or trim conversations and steps that are not of interest prior to aggregation; (2) **Path simplification** - users can reduce the variance of the aggregation by removing duplicate consecutive steps; (3) **Interaction** - users can perform several interactions, such as click to expand the next level of sequences; click to select; zoom & pan; and hover to view statistics on a tooltip. (4) **Reverse journey** - users can order and aggregate conversations from end to start, allowing an analyst to observe which conversation steps lead to an undesirable end of the conversation. (5) **Trend analysis** - users can display a comparison chart that computes the increase/decrease in volume or rates between two periods or two groups of conversations. A version of the flow analysis has been released by the authors as open source to the Watson Assistant public github repository [6].

5 PROJECTS AND DATASETS

We were involved in about a dozen client projects that used our methodology and tool. Evaluation examples presented in Section 6 are based on the data of the AskHR chatbot, which was designed to support employees' and managers' questions and transactions in HR processes such as payroll, employee management, and travel.

This bot was deployed on both web and Slack channels and handles more than 170k conversations per month, with a monthly increase in conversation volume and user adoption. The bot contains around 30 skills to cover specific topic areas, and an orchestrator skill that routes the conversation to the right content skill. Our involvement in this project lasted for the last 12 months on a weekly basis, where we deployed and validated our tools and provided around 20 recommendations that were adopted by the analysts team.

In another significant client project, we partnered with a large national healthcare organization that provisioned COVID-19 vaccines and PCR tests. This bot was deployed in five different channels of call centers, and the daily call volume was larger than 200k with frequent capability and content updates on a weekly basis. During the three months of our involvement, we helped the client detect numerous problems related to missing intents and entities, training

²In machine learning, feature interaction refers to the joint effect of two or more features on the prediction

existing intents, and responding to previously mentioned entities in a single conversation. Due to confidentiality issues, we cannot provide more specific information and examples about this project.

6 DEMONSTRATION AND INITIAL ASSESSMENT

In this section, we present a couple of issues that were found in the AskHR project and demonstrate initial insights and evaluation from using the root cause and path flow analysis capabilities.

6.1 User struggle at slot filling dialog nodes

Timeoff Request is a frequently-used process task in the AskHR chat-bot. However, its completion rate was only around 50% in December 2020, so the analysts wanted to understand which conversation dialog turns were mostly related to failures and try to improve those. In Watson Assistant, this task is implemented using a feature called *slot filling*, which handles the various ways of collecting input entities, such as the number of timeoff days or hours, without explicitly programming their execution logic. The turn labels represent the names of dialog nodes that correspond to slot-filling logic. The analyst can then configure the analysis to investigate patterns related to conversation turns.



Figure 4: Dialog node sequences related to failure of *Timeoff Request* task

Figure 4 shows the result of the root cause analysis applied in this issue. The three most significant patterns of consecutive turns were detected and presented. The first pattern relates to *ERROR_SOAP* runtime error code that appears when the user intent can't be fulfilled by the backend service. The second pattern represents a user stuck at the *provide_date* step for two consecutive turns. The red bar represents 22% of the failed conversations that exhibited this sequential turn pattern. To gain trust in the results, the blue bar representing success of only 2% is shown. This view helped the analyst detect this issue and estimate the potential volume and impact of fixing it for improvement.

To understand the potential cause, the analyst clicked on this pattern and explored a summary of the user inputs using an N-gram analysis that displays most frequent words and phrases in the selected set of utterances. It turned out that *time balance* appeared frequently at that location of the dialog. Then, with additional drill-downs into a few transcripts, the analyst was able to determine that the bot didn't correctly understand the timeoff balance request and the different ways that users asked it. After the problem was understood, an option was added to allow users to review their timeoff balance before or during the timeoff request. After deploying these changes, the task completion rate improved by 10%.

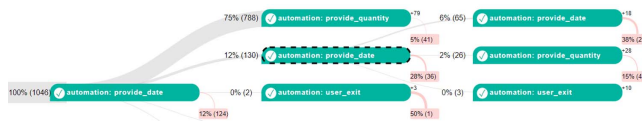


Figure 5: User struggle in *Timeoff Request* task. Next-step flow distributions and user abandonment can be observed

Figure 5 shows the results of applying the path flow analysis on the same issue. The flow graph was filtered to focus on conversations that reach the *provide_date* node and their next-step distribution and outcomes. Here the analyst was able to observe additional characteristics of this issue, e.g., that 12% of conversations were abandoned after reaching that node, another 12% were stuck for two turns, and 65 conversations were stuck for 3 consecutive turns.

6.2 Escalation to live human agent chat

In many cases, escalations to chat with a live human agent are considered an undesirable outcome. Figure 6 shows the results from analyzing this phenomenon via the root cause analysis of user inputs. Escalation was selected by the analyst for the failing group and conversations that received positive feedback were used for the comparison. The algorithm discovered five feature values that correspond to two topic areas: *income tax*, *tax deduction* and *proof submission* are related to tax issues, while the other two refer to requests of HR contact phone numbers. Through additional transcript drill-downs, the analyst found two new topics that the bot didn't handle or where it gave too generic response.

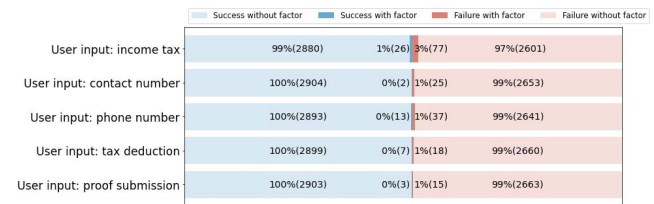


Figure 6: Bigrams in user inputs leading to escalation

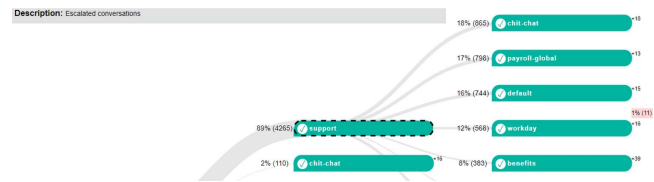


Figure 7: Reverse path flow chart showing skills that lead to escalation

The analyst also wanted to know which skills lead to escalation, and used the reverse path flow analysis option, which shows which skills lead to a certain outcome. In Figure 7, we can observe that 89% of conversations escalate from the *support skill*, which is a skill responsible for escalations, and *Payroll*, *Workday*, and *Benefits* are the top HR tasks that lead to escalations. This allows the analyst to focus on improving these content skills.

7 DISCUSSION AND NEXT STEPS

The client teams we worked with made dozens of improvements based on insights from the tool and gained more confidence in understanding issues and assessing improvement options. Our main conclusion for improvement would be additional investments in the development of an even more intuitive UI. This could help analysts more easily understand why we're calling their attention to specific features or phenomena and estimate the potential for improvement.

Overall, these are promising results and we plan to continue to enhance it. We also plan to perform formal validations by incorporating capabilities for client feedback into the tool and measuring effectiveness and false-positive rates for presented insights.

In root cause analysis, we plan to take client preferences and history into account. Features that are considered non-informative will be excluded from the feature pool and not used in the future runs. We're also considering pursuing research that distinguishes between correlated and causal factors of conversation failures. In path flow analysis, we plan to adapt it to the new Action skill model of Watson Assistant [4].

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